

Version 3.0 — Jan 2026

“Seamlessly Connecting Housing, Transportation, and More Through Blockchain.”

1. Abstract

BIG serves as the backbone of a dynamic Web Platform called KeyChain designed to host decentralized applications that power everyday life. Its mission is simple but transformative — to bring traditional Web2 transactions into the Web3 economy through seamless, transparent, and reward-driven blockchain technology.

The KeyChain ecosystem will evolve through **multiple open-ended phases**, each expanding the real-world use cases of the BIG token. Phase I, spanning from the present through December 31, 2027, is dedicated to transforming real-world transactions by implementing ERC-7943 and ERC-3643 standards via the LeaseFi decentralized application (dApp).

LeaseFi enables property buyers, sellers, landlords, and tenants to interact through **smart contracts**, ensuring automated, transparent, and secure agreements without intermediaries. With built-in **staking incentives** and **transaction rewards**, BIG delivers tangible value while creating an interconnected token economy that rewards users for simply living their daily routines.

By anchoring real-world commerce — from property deals to future transportation and lifestyle applications — to a single token framework, BIG aims to redefine how people pay, earn, and interact in the digital economy.

2. Introduction / Problem Statement

Modern digital transactions remain fragmented, expensive, and opaque. Whether in real estate, transportation, or everyday services, users face **high fees**, **low accountability**, and a **lack of interoperability** between the tools they rely on. Most online platforms operate in isolation, limiting user choice, creating redundant processes, and often obscuring where costs actually go.

Transparency is another major issue. In many industries, customers have little visibility into **which services are being provided**, **what steps are required to complete a transaction**, or **which vendors are available**. This lack of clarity leads to inefficiency, mistrust, and wasted resources on both sides of the transaction.

Take real estate, for example. A first-time buyer or seller often enters the process without a clear understanding of what is required — from inspections and appraisals to title services and closing costs. The absence of a unified, trustworthy platform leaves individuals vulnerable to misinformation, hidden fees, and unnecessary intermediaries.

3. Proposed Solution

KeyChain aims to bridge this knowledge and accountability gap by providing an open, blockchain-powered marketplace where all participants can see the **required steps**, **qualified service providers**, and **associated costs** in real time. Through smart contracts and verifiable data, KeyChain establishes a new standard of **transparency**, **interoperability**, and **fairness** — ensuring that users always understand what they're paying for and why.

BIG powers KeyChain, a unified mobile and desktop dApp:

LeaseFi - Blockchain Real Estate Transaction Layer on KeyChain dApp Enabling:

- Smart rental agreements
- Automated rent payments
- Escrow protection for deposits
- On-chain tenant credit scoring
- Efficient, cross-border rental management for landlords
- Fractionalized property ownership for investors
- Clean, deduplicated MLS data with no clutter
- Instant payment options (crypto, Apple/Google Pay, bank rails)
- On-chain notarization for deposits and offers
- Direct off-market offers available to verified buyers with proof of funds

Pilot Market & Expansion:

(Details retained: Hampton Roads → Charlotte → Florida.)

4. Tokenomics

4.1 Token Overview

- Token Name: BIG
- Symbol: BIG
- Blockchain: Casper Network (CEP-18 standard)
- Max Supply: 5,000,000,000 BIG
- Decimals: 18

4.2 Utility & Use Cases

BIG powers the ecosystem by enabling:

- Payments for services on the KeyChain platform
- Staking for passive rewards
- Early access to premium features
- Rewards for referrals, usage, and community engagement

4.3 Distribution

Category | Allocation | Vesting Details

---|---|---

Team & Founders | 15% | 12-month cliff, 48-month vest

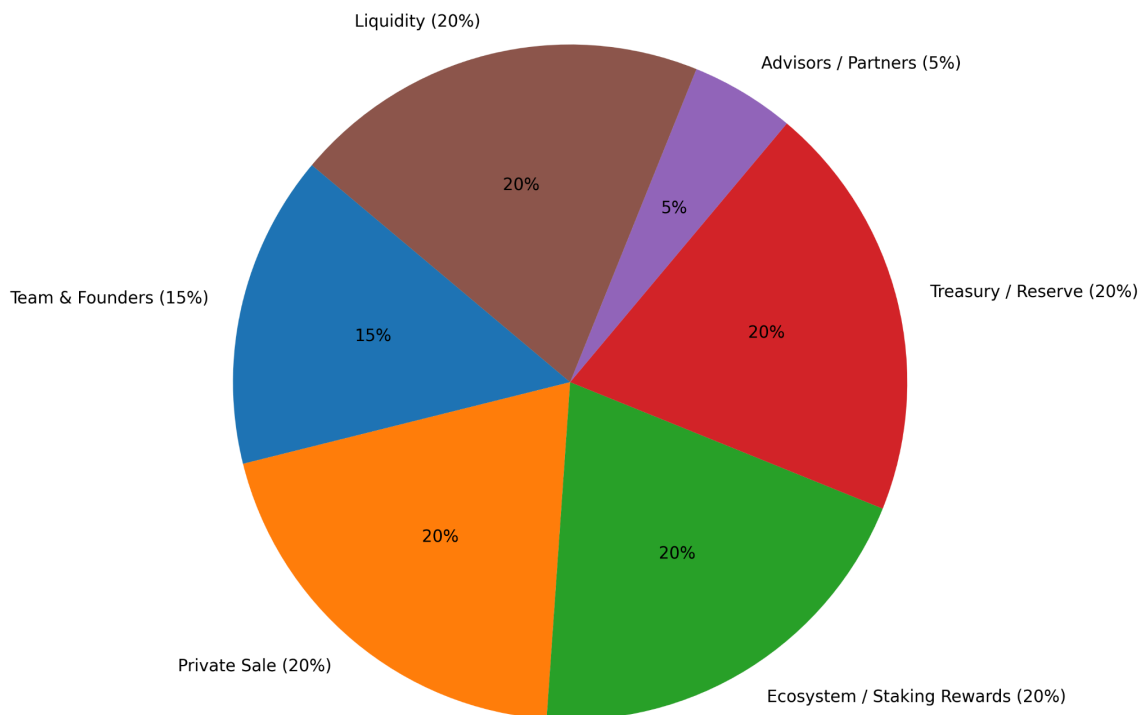
Private Sale | 20% | 6-month cliff, 12-month vest

Ecosystem/Staking Rewards (Staking Reserve Pool)| 20% | Released at 6% annually

Treasury / Reserve | 20% | Using for Future Phased projects

Advisors / Partners | 5% | 12-month cliff, 48-month vest

Liquidity | 20% | Immediate for DEX/CEX pools



4.4 Vesting & Lock-Up

- Team, founders, advisors, and partners: 12-month cliff, then 48-month linear vesting automatically vested into the staking pool and locked

- Reduces early sell-off and ensures long-term commitment

4.5 Inflation / Deflation Policy

- Supply: Fixed at 5B BIG
- Rewards: Fixed reserve staking pool

4.6 Pricing & Fundraising

- Initial Price: \$0.0015 per BIG
- Hard Cap: \$1,500,000

Funds Allocation:

- Development: 35%
- Marketing: 25%
- Operations: 20%
- Legal & Compliance: 10%
- Reserve: 10%

4.7 Growth-Oriented Token Fundamentals

The BIG token is designed as a growth-focused asset rather than a traditional dividend-style instrument. Using a dynamic scheduled token release provides staking yield opportunities, but the primary value driver lies long-term in ecosystem expansion and transaction volume.

Why Growth over Dividends?

- Reinvestment Priority: BIG prioritizes channeling resources into infrastructure, R&D, and marketing—fuels the accelerated adoption of KeyChain.
- Staker Rewards as Bonus: Staking payouts exist but scale gradually in favor of committed long-term holders rather than high short-term dividends.
- Network Effect Model: Greater utility and adoption drive transaction volume, which in turn drives staking yield and overall value.

Proven Growth-First Models

This mirrors the strategies of leading global firms:

- Nvidia & Amazon: Reinvested profits into R&D and infrastructure, prioritizing dominance over near-term payouts.
- Google & Microsoft: Built expansive ecosystems through sustained reinvestment, only later adding modest dividends once growth matured.
- Costco: Maintained a low-payout model while reinvesting in logistics, technology, and customer growth—driving massive long-term returns.

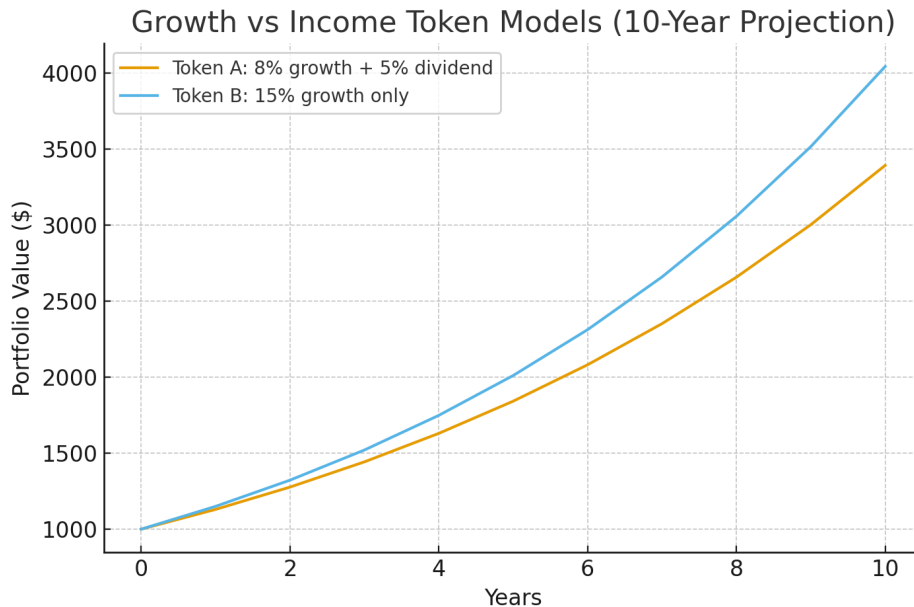
Appeal for Investors

- Exposure to a scalable ecosystem that spans transportation, rentals, and real estate listings.
- Alignment with long-term appreciation instead of short-term yield.
- Staking multipliers reward loyalty, while dynamic fee allocation ensures that as adoption

stabilizes, committed holders gain more of the revenue stream.

- Transparent on-chain metrics reinforce trust.

Comparison: Growth vs Income Model



- Token A (8% growth + 5% dividend): Resembles a dividend stock—steady returns but slower appreciation.

- Token B (15% growth only): Resembles BIG’s design—high reinvestment, strong compounding, greater upside over 10+ years.

BIG leans toward Token B: growth-oriented, with reinvestment mechanics driving value, while staking provides supplementary yield rather than being the core value proposition.

4.8 Transaction Fee Allocation

A 2% transaction fee is applied to all transactions conducted within the KeyChain ecosystem (LeaseFi and future modules). This fee is processed via smart contracts and distributed via the BIG dynamic scaling model.

Dynamic Scaling Model:

BIG utilizes a capped 6% annual release sourced exclusively from its 20% Staking Reserve Pool, issuing approximately 60 million tokens per year over ~17 years, after which it fully ceases and staking rewards transition to a fee-only model.

LeaseFi Staking Yield Model – 6% Inflation

Staked Supply (%)	Fee Allocation to Stakers	Total Nominal APR	Compounded APY	Annual Earnings (per 1M)	Compounded Earnings (per 1M)
≤ 30%	0%	20.00%	≈ 22.14%	200,000	221,373
≤ 40%	20%	16.00%	≈ 17.59%	160,000	175,938
≤ 50%	40%	13.60%	≈ 14.55%	136,000	145,477
≤ 60%	60%	12.00%	≈ 12.75%	120,000	127,480
≤ 70%	80%	10.86%	≈ 11.47%	108,571	114,698
≤ 80%	100%	10.00%	≈ 10.52%	100,000	105,156

Under this design, stakers earn rewards from two sources: tokens released from the Staking Reserve Pool Supply and transaction fees. Annually, the protocol releases 6 % of the Staking Reserve Pool Supply and distributes it pro rata to stakers, so the driven return per staked token equals 6 % divided by the fraction of tokens that are bonded. Because only staked tokens participate, lower participation produces higher per-token rewards.

Value Alignment:

- Long-term stakers benefit as adoption stabilizes
- Rewards will heavily incentivize users of the KeyChain Ecosystem
- All holders profit from transaction volume

BIG Staking Reserve Pool Token Release Schedule

Core Assumptions (explicit for token holders)

- Total BIG Supply: 5,000,000,000
- Staking Reserve Pool Allocation: 20% = 1,000,000,000 BIG
- Release Rate: 6% annually
- Releasing Source: Staking Reserve Pool only
- Releasing Method: Fixed annual issuance until the Staking Reserve Pool is exhausted
- Distribution: Tokens flow into the protocol (staking rewards and treasury per governance rules)

Annual Staking Reserve Pool Release Math

- Annual release amount:
 $1,000,000,000 \times 6\% = 60,000,000$ BIG per year
- Staking Reserve Pool duration:
 $1,000,000,000 \div 60,000,000 \approx 16.67$ years
- Practically: 16 full years of issuance + a final partial year

Why This Releasing Model Is Made For Long-term Holders

1. No runaway inflation

- Inflation is capped by the Staking Reserve Pool
- No perpetual minting
- No surprise dilution

2. Long runway (≈ 17 years)

- Supports:
 - Long-term staking rewards
 - Ecosystem growth
 - Fee-based transition to sustainability

3. Smooth, predictable supply curve

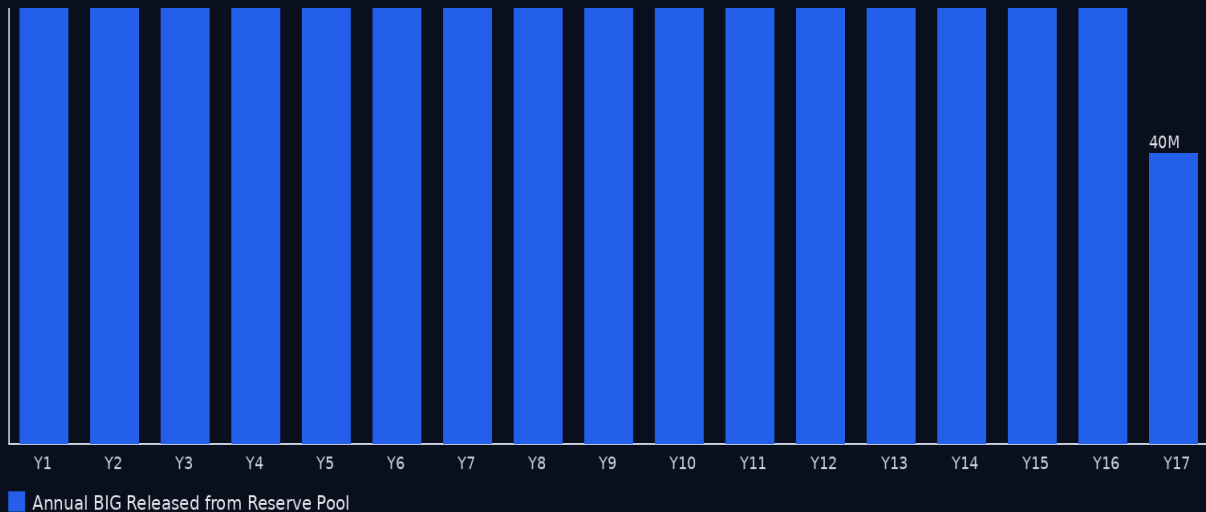
- Fixed annual issuance
- Easy to model
- Easy to audit on-chain

4. Clean transition to fee-only economics

- Once the Staking Reserve Pool is exhausted:
- Inflation $\rightarrow 0$
- Staking rewards become 100% fee-driven
- Aligns perfectly with your Dynamic Fee Allocation Model

BIG Reserve Pool Supply — 6% Annual Release Schedule

Reserve Pool: 1,000,000,000 BIG • Annual Release: 60M BIG (Final Year: 40M)



4.10 Pre-Initial Coin Offering Funding

The **Pre-ICO phase** will allocate **20% of the total token supply (1,000,000,000 BIG tokens)** at an initial price of **\$0.0015 per BIG**, aiming to raise **\$1,500,000 USD**.

Funds from this round will support **application upkeep, maintenance, and continued development** over 24 months.

The **Pre-ICO sale** will remain open for **7 days** or until the allocation is fully subscribed. Upon completion, **any unsold tokens** will be **redirected to the liquidity pool** to strengthen market depth and post-launch stability.

All proceeds will be used exclusively for project development and operational sustainability according to the following funding plan:

Pre-ICO Funding Plan: There will be three tiers of priority hires, based on pre-ICO funding. Funds will be used to pay onboarding members for 24 months. Any amount of funding that isn't obtained through the Pre-ICO will be obtained through transaction fees.

4.11 Priority Hires

Role	Key Responsibilities (brief)	Evidence/Notes (brief)
Product Manager (PM)	Defines product roadmap; prioritizes features; coordinates across teams	Ensures the platform meets landlord, tenant and manager needs
Project Manager/Scrum Master	Manages timelines and sprints; facilitates communication; tracks deliverables	Keeps development on schedule
Blockchain Developer/Smart-Contract Engineer	Develops smart contracts for leases, rent payments and maintenance; implements tokenization	Smart contracts can automate rent collection and lease agreements
Backend Developers	Build APIs; integrate payment, identity and data services; manage databases	Ensure secure and scalable server-side architecture
Frontend/Full-Stack Developers	Develop user interfaces for web/mobile; implement portals and dashboards	Deliver seamless UX across tenant, landlord and manager portals

Role	Key Responsibilities (brief)	Evidence/Notes (brief)
QA/Test Engineers	Design automated/manual tests; verify smart-contract behaviour; conduct security and performance testing	Ensures reliability and compliance
UI/UX Designer	Design intuitive user flows; ensure accessibility and brand consistency	Improves adoption among non-technical users
Data Engineer / Data Scientist	Build data pipelines; develop analytics and AI/ML models	Support data-driven decision making

(credit scoring, predictive maintenance)

4.11.1 Secondary Hires

Role	Key Responsibilities (brief)	Evidence/Notes (brief)
Compliance & Risk Officer	Develop policies (data protection, AML/KYC, Fair Housing); conduct audits; manage risk	Coordinates with legal and tech teams to ensure regulatory compliance
Security Engineer	Implement encryption and key management; enforce secure coding practices; monitor for breaches	Protects user data and blockchain assets
Tenant Relations & Support Manager	Oversee customer support; handle tenant issues; mediate disputes	Ensures positive tenant experience

Role	Key Responsibilities (brief)	Evidence/Notes (brief)
Maintenance & Vendor Coordinator	Build vendor network; schedule repairs and preventive maintenance; track vendor performance	Ensures timely resolution of maintenance requests
General Counsel/Real-Estate Attorney	Draft and review contracts; monitor legal changes; manage tenant disputes and evictions; ensure Fair Housing compliance	Attorneys help navigate complex regulations and draft enforceable contracts and handle disputes/evictions

Insurance Specialist	Arrange liability, property and cyber-insurance coverage; review policies	Mitigates financial exposure from incidents
Chief Financial Officer (CFO)	Oversee budgeting, forecasting and investor reporting; manage treasury	Aligns financial planning with growth
Certified Public Accountant (CPA)	Handle accounting and tax compliance; track income/expenses; prepare statements and plan taxes	CPAs navigate complex rental property tax laws, maintain accurate bookkeeping and maximize deductions
Accounts Receivable/Payable Specialist	Manage rent collection; process vendor payments and security deposit refunds	Reconciles on-chain and off-chain transactions
Investor Relations Manager	Communicate with investors; prepare financial updates; manage fractional ownership/tokenization communications	Needed if offering tokenized real-estate assets
Role	Key Responsibilities (brief)	Evidence/Notes (brief)
Marketing Manager & Growth Hacker	Develop brand strategy; manage digital campaigns; drive tenant and landlord acquisition	Balances B2C and B2B marketing
Sales & Business Development	Form partnerships with property owners, investors and service providers; negotiate deals	Expands business reach

Community Manager	Engage users via social media and forums; collect feedback; foster community	Supports user retention and network effects
Customer Support Agents	Provide 24/7 assistance via chat, email and phone; handle maintenance tickets and payment issues	Ensures responsive customer service
Human Resources Manager	Recruit and onboard staff; manage training; ensure HR compliance	Fosters company culture and compliance
Office Manager/Administrative Assistant	Handle day-to-day administrative tasks; coordinate schedules and procurement	Keeps operations running smoothly
Optional Roles (AI/ML Engineer, Tokenomics Consultant, Real-Estate Valuation Analyst)	Develop predictive models; design token incentives; analyze property values	Enhance advanced functionality and innovation

5. How It Works - KeyChain App Ecosystem

KeyChain is the central mobile and desktop dApp that unifies core platforms into a single, intuitive user interface.

Real Estate on the Blockchain

LeaseFi allows users to buy, sell, and rent properties directly in the app using fiat or cUSD on the Casper network. It offers fractionalized ownership, customizable smart contracts, and a three-tab interface: Buy, Sell, Rent. Transaction fees are converted into BIG tokens and distributed according to the platform's allocation model.

LeaseFi Rental Platform – Proposed System Architecture

Background

Property-management platforms typically combine front-end portals for tenants, landlords, and vendors with back-end services to handle leases, payments, maintenance, and other tasks. A recent guide on developing custom property-management software emphasises that these systems commonly follow a **three-tier architecture** with a **user interface (UI)** layer, an **application layer** for business logic, and a **database layer** for storing data. Tenant portals can allow renters to view the status of their lease, rent amount, deposits, and submit maintenance requests. Maintenance request tracking is highlighted as a key module because it saves time for tenants and landlords by improving collaboration and accelerating repairs. Residential software also needs to handle the signing of lease agreements, receive rent payments, and track occupancies; commercial systems additionally track taxes, insurance, and maintenance fees.

LeaseFi appears to be positioned as a **smart rental payment platform** that performs KYC/AML checks and uses AI to score credit. LeaseFi intends to expand beyond payment processing to provide lease management, maintenance transactions, and rental-insurance services. Following agile best practices, it is sensible to start with a **Minimum Viable Product (MVP)** focusing on rental lease agreements, property management, and then iteratively add maintenance requests and insurance.

Architecture overview

The proposed system architecture (see diagram below) adopts a **modular microservice-based design** to support LeaseFi's planned growth. Key components include:

1 User interface layer

- **Web / Mobile UI (User Portal)** – A responsive portal used by tenants, landlords, maintenance vendors, insurance providers, and administrators. Users can sign up, authenticate (using multi-factor if needed) and manage their profiles.
- **Role-specific dashboards** – Tenants view lease status, rent amount, and submit maintenance requests; landlords manage properties, view occupancy, and monitor

payments; vendors receive work orders; insurance partners manage policies.

2 API gateway and identity management

- **API Gateway** – Receives all requests from the UI, applies rate-limiting, logging, and forwards calls to appropriate microservices. The gateway enables versioning and enforces security policies.
- **Identity and Access Management (IAM)** – Handles user authentication/authorization, implements KYC/AML checks, and credit-scoring using AI models. This enables single sign-on across all services.

3 Service layer (microservices)

- **Lease Agreement Service** – Provides functionality to create, negotiate and sign digital rental contracts. Documents are stored in the Document Storage and hashed onto the Casper blockchain for tamper-proof verification. The service integrates with the **Payment & Accounting** service to collect deposits and monthly rents. LeaseFi also exposes a **rent reporting** API so landlords can view payment history.
- **Maintenance Service** – Manages maintenance requests from tenants, records work orders, assigns vendors, and tracks resolution status. Maintenance events and costs are written to both the relational database and the BIG ledger to create an immutable log. The module supports scheduling, image uploads, and real-time notifications. Automatic reminders are sent via the **Notification & Messaging** service.
- **Insurance Service** – Facilitates rental insurance policies and liability coverage. Tenants can purchase policies through the UI; the service integrates with external insurance providers via API and tracks policy status. Insurance data is recorded on the ledger for transparency. When a claim is lodged (e.g., due to damage or liability), the service can reference maintenance and lease data.
- **Payment & Accounting Service** – Processes rent, deposit, and insurance payments. LeaseFi interfaces with external payment gateways to collect funds and uses an accounting engine to create receivable/payable entries, perform automatic

bank reconciliation and generate financial reports.

- **KYC & Risk Scoring Service** – Runs identity verification and credit-scoring algorithms when tenants apply for leases or insurance. LeaseFi uses AI models to compute risk profiles and shares results with other services.
- **Notification & Messaging Service** – Provides email/SMS/push notifications, in-app chat, and automated reminders. LeaseFi ensures all stakeholders receive updates (e.g., lease renewal reminders, maintenance status changes).

4 Data layer

- **Relational Database & Document Storage** – Stores structured data about users, properties, leases, maintenance tickets, and payments. LeaseFi also holds PDF contracts and image attachments. Backups and encryption protect sensitive data.
- **BIG Blockchain Ledger** – Every lease agreement, maintenance transaction and insurance policy is represented by a smart contract on the BIG ledger. Hashes of documents and records are stored on-chain to guarantee integrity and provide an auditable history. The front-end UI can automate payment schedules and push transactions to Smart contracts (e.g., release of deposits) and enable decentralized approvals.

5 External integrations

- **Payment Gateway** – Forwards transactions to banking networks and credit-card processors. It ensures timely rent and premium collection and returns payment status to the Payment & Accounting service.
- **Insurance Provider API** – Interfaces with third-party insurance platforms to underwrite policies, verify coverage, and handle claims.
- **Vendor Management** – Integrates with contractor directories or CRM systems to source maintenance vendors, manage their profiles, and track service quality.

6 Security & Compliance

The architecture employs multiple layers of security: API gateway with rate-limiting and TLS, IAM enforcing granular roles, encryption of data at rest, and compliance with GDPR/CCPA and local regulations (e.g., deposit handling rules). Storing transaction hashes on the blockchain enhances data integrity.

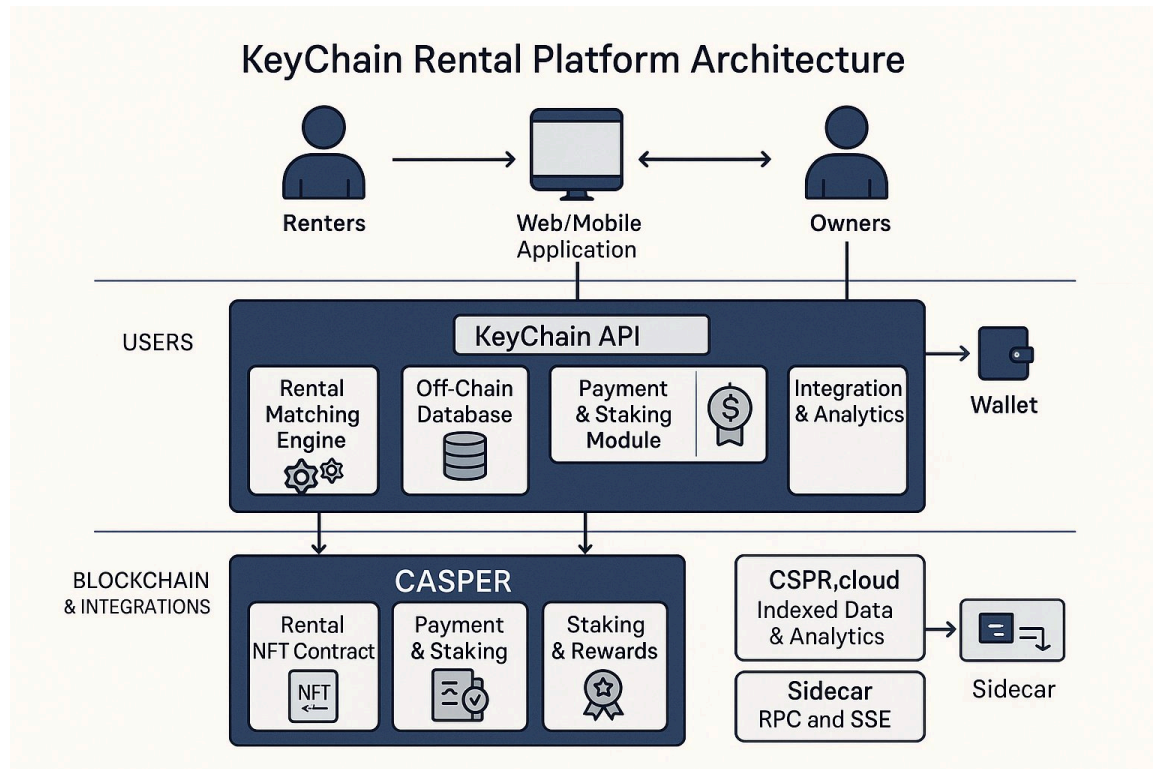
Iterative development strategy

1. **Phase 1 – Lease Agreements MVP:** focus on the Lease Agreement Service, Payment & Accounting, and Property Management. Implement digital contracts, rent collection, and deposit handling. Deploy the blockchain ledger to record contract hashes and payment events.
2. **Phase 2 – Maintenance Requests:** build the Maintenance Service and Notification module. Extend the UI for maintenance submission and vendor assignment. Integrate with external vendor management as needed.
3. **Phase 3 – Insurance Module:** add the Insurance Service and connectors to insurance providers. Implement policy purchase and tracking. Use on-chain smart contracts to record policies and claims.

This phased approach aligns with agile recommendations to avoid developing every feature at once and instead prioritize high-value functionality.

Architecture diagram

Below is the proposed system architecture for the KeyChain rental platform. Users interact with the Web/Mobile portal, which communicates through an API gateway to modular microservices. Services access both the relational database and the KeyChain blockchain ledger for data storage and immutability. External payment, insurance, and vendor systems integrate via well-defined APIs.



6. Staking & Rewards Model

To strengthen community participation and reward early adopters, BIG features a transparent staking and rewards framework tied directly to on-chain transaction activity.

Transaction Fee Distribution

A **2% transaction fee** is applied to every transaction conducted within the LeaseFi platform and, subsequently, across all KeyChain ecosystem applications. These fees are automatically processed through smart contracts and distributed according to the

Dynamic Scaling Fee Allocation Model

The dynamic model ensures equitable growth between user incentives and staker rewards as the ecosystem matures. This schedule uses participation tiers:

- If 30 % or less of the total supply is staked, 100 % of the 2 % fee goes to the treasury and stakers get no fees.
- Above 30 % and up to 40 % staked, stakers receive 20 % of the fee (0.4 % of staking reserve pool supply) while 80 % goes to the treasury. The nominal APR is $0.4 \% \div 0.40 = 1.0 \%$.

- At 50 % staked, stakers receive 40 % of the fee (0.8 % of rewards pool supply); nominal APR is $0.8 \% \div 0.50 = 1.6 \%$.
- At 60 % staked, they receive 60 % of the fee (1.2 % of rewards pool supply); nominal APR is $1.2 \% \div 0.60 = 2.0 \%$.
- At 70 % staked, 80 % of the fee goes to stakers (1.6 % of rewards pool supply); nominal APR is $1.6 \% \div 0.70 \approx 2.2857 \%$.
- At 80 % staked, stakers capture the entire 2 % fee; nominal APR is $2 \% \div 0.80 = 2.5 \%$.

This gradual scaling model increases the share allocated to long-term stakers as adoption stabilizes and transaction volume grows.

Proportional Staking Rewards

When fees are directed to the staking pool, they are distributed proportionally according to each participant's share of the total staked BIG tokens.

For example:

- If a wallet holds **10% of all tokens** currently staked, it receives **10% of all staking pool rewards** distributed during that period.

This ensures fairness and direct alignment between staking participation and reward yield.

Unbonding Period

To maintain staking stability and prevent rapid withdrawal manipulation, all staked BIG tokens are subject to a **48-hour unbonding period**. During this time, the tokens remain locked in the staking contract and cannot earn rewards or be withdrawn until the unbonding window concludes.

Early Transaction Rewards for Pre-Sale Participants

Beginning in **December 2026**, transactions processed through the **LeaseFi application** will start generating 2% fees. These fees will accumulate within the staking contract, forming the initial liquidity and rewards foundation for the network.

This model allows early investors to benefit immediately from the platform's live transaction activity and rewards those who supported the ecosystem before the ICO launch.

Realtor Rewards & Ranking System

LeaseFi introduces a competitive rewards framework for realtors and agents, modeled after performance-based scoring. Agents compete for rewards based not only on total sales but also on efficiency, consistency, and reputation. Rankings are continuously updated with blockchain-anchored transparency.

Ranking Categories:

1. Transaction Volume & Sales Performance

- Closed Transactions (# of deals)
- Total Sales Volume (\$ value)
- Average Deal Size
- Days on Market (DOM)

2. Listings & Inventory Management

- Number of Active Listings
- List-to-Sale Price Ratio
- Expired/Withdrawn Listings

3. Client & Market Reach

- Buyer vs. Seller Representation Ratio
- Geographic Market Share
- Lead Conversion Rate
- Referral/Repeat Business Rate

4. Marketing & Visibility

- Online Presence & Engagement
- Photography & Media Quality
- Ad Spend ROI

5. Professionalism & Client Satisfaction

- Customer Reviews & Ratings
- Net Promoter Score (NPS)
- Communication Speed
- Negotiation Skills

Realtor Rankings: Agent Quality Index (AQI)

We introduce an objective, tamper-resistant ranking framework for real estate agents on LeaseFi. Scores are computed per segment (e.g., starter homes, luxury, rentals) and globally (volume-weighted across segments).

Scoring Framework

All underlying metrics are winsorized (5th–95th percentile) and converted to 0–100 via within-segment percentiles.

Default Weight Schema (admin-adjustable):

- Production — 35%
- Efficiency & Speed — 25%
- Client Experience — 20%
- Compliance & Reliability — 15%
- Marketing Quality — 5%

Formula:

$$\text{AQI} = \sum (\text{weight}_i \times \text{score}_i).$$

Versioning:

Maintain a versioned “weight schema” so historical AQI calculations remain reproducible across schema changes.

Deal Difficulty Adjustment (Skill, Not Luck)

We layer on a Bayesian “skill delta” per transaction to reflect performance versus expectations on varying difficulty levels.

Train a supervised model to estimate expected Days-to-Close (DTC) and expected sale/discount given features such as property attributes, seasonality, lender type, financing terms, inspection flags, and local inventory.

Compute residuals: $\text{residual} = \text{actual} - \text{expected}$.

Update an Elo/TrueSkill-like rating per agent: outperforming expectations on harder deals (e.g., complex contingencies, low inventory) yields larger rating gains than routine wins.

Anti-Gaming & Integrity Controls

Verified closings only: anchored by on-chain escrow/recording, MLS status change, and title settlement record hash.

Review hygiene: exactly one review per transaction ID; anomaly detection for bursts and IP/device duplication; unverifiable reviews are down-weighted.

Lead attribution sanity: first-touch + last-touch with time-decay; penalize “lead hoarding” without activity.

Pricing accuracy discipline: serial micro-cuts beyond segment norms nudge Pricing Accuracy downward.

Duplicate listing scrubs: property fingerprint (normalized address + geocode + unit + lot + living_area + parcel_id → SHA-256).

Segment primacy: global rank is visible, but segment ranks are primary to prevent “luxury sniping” from dominating starter-home boards.

Computation & Aggregation

For each agent and segment, compute: (1) normalized metric scores (0–100), (2) weighted AQI, and (3) skill-adjusted AQI using the Bayesian delta. Global AQI is then a volume-weighted aggregation across segments with the current weight schema attached as metadata.

All inputs, intermediate outputs (winsorization bounds, percentile bins), and final scores are logged with content-addressed hashes. This ensures auditability, reproducibility, and dispute resolution through cryptographic proofs anchored on-chain.

6. Career Growth & Credentials

- Experience Level (years licensed)
- Certifications & Specialties
- Brokerage Rankings

Functional Overview:

- LeaseFi aggregates these metrics to create leaderboards that update in near real-time.
- Agents earn performance rewards in BIG tokens, distributed through the rewards pool.
- AI-driven dashboards provide real-time feedback, similar to how Uber ranks drivers.
- Top performers gain visibility on the platform, priority placement in search results, and additional staking multipliers.

Why It Matters:

- For Agents: Continuous performance tracking helps improve skills and earn rewards.
- For Clients: Verified, ranked agents reduce risk and increase trust.
- For the Ecosystem: Introduces a competitive, gamified environment that elevates professional standards and accelerates adoption of KeyChain.

7. Roadmap

Q1 2026 – Pre-Sale / Pre-ICO Stage

- **Pre-Sale for \$BIG token** to early investors and platform participants.
- **Begin marketing and community onboarding**
- **Audit all smart contracts and staking pool mechanisms before ICO.**

Q1–Q4 2026-2027 – Infrastructure & Revenue Activation

Q3–Q4 2026 – Development & Testing

- **Build Smart Lease Builder** to automate lease creation and signing using Casper smart contracts.
- **Integrate Escrow Contracts** to manage deposits, rent payments, and revenue disbursements.
- **Test staking and rent automation** modules in closed beta.
- **Develop initial transaction fee logic** (2% per contract execution).

Q4 2026 – Foundational Launch

- **KeyChain Website & Landlord Tools Launch**
 - Public release of LeaseFi's core web platform.
 - Launch of **Landlord Dashboard**, including property listings, tenant management, and maintenance tracking tools.
 - **Begin data aggregation** for property metrics & rental yield analytics.
 - Establish **backend architecture** for smart contract integration (Casper Network base).

Q4 2026 – Fiat Integration & Global Expansion

- **Implement multi-currency fiat on/off ramps** (USD, SGD, BHD).
- **Expand LeaseFi operations into EU and MENA markets**, localized for housing and tax compliance.
- **Launch white-label program** for real-estate firms.
- **Integrate ISO 20022 messaging compatibility** for global financial interoperability.

Q2 2027 – Platform Monetization & Global Pilot

- **Smart Lease Builder + Escrow Launch (January 2027)**
 - Live deployment of smart contract
- **Rent Automation + Staking Integration (January 2027)**
 - Every transaction automatically channels 2% platform fee
 - Real-time reward tracking begins for early adopters.
- **Market Launch Focus:**

- Target initial markets: **United States, Singapore, and Bahrain** for regulatory diversity and pilot scalability.

Q3 2027 – ICO & Tokenized Revenue Distribution

- **Public ICO Launch for \$BIG token.**
 - Distribute tokens to investors and list on centralized and decentralized exchanges.
- Introduce **investor dashboard** for real-time visibility into revenue share and staking performance.
- Begin **integration of on-chain KYC/AML** features in compliance with ERC-3643 and 7943.

Q1 2028 – Credit Scoring & Tax Infrastructure

- Launch **AI-Driven On-Chain Credit Scoring** system, building on historical rent payment data and smart contract behavior.
- Begin development of **Tax Preparation Module** for CPAs:
 - Enables secure data pulls from clients' property income and expenses.
 - Generates compliant, exportable tax summaries (U.S. and Singapore standards).

8. Legal Disclaimer

This document is for informational purposes only and does not constitute an offer to sell, a solicitation to buy, or a recommendation for any security or cryptocurrency. Participation in the BIG ecosystem involves risks, including market volatility, regulatory changes, and technology failures.

Conclusion

By adopting a modular microservices architecture with a blockchain ledger, the KeyChain platform can evolve from a smart rental payment solution into a full-fledged real-world ecosystem. Starting with digital lease agreements and gradually layering maintenance and insurance modules allows the platform to deliver value quickly while maintaining a flexible foundation for future enhancements.